**Netflix Data Analysis and Visualization Project**

**1. Project Overview**

This project focuses on analyzing the Netflix Titles dataset to understand trends in movies and TV shows available on the platform. The objective of this analysis is to explore the structure of Netflix content, identify patterns in genres, analyze the distribution of movies and TV shows, and understand how Netflix content has grown over time.

The dataset was cleaned and processed using Python libraries such as Pandas in a Jupyter Lab environment. After performing data cleaning, transformation, and analysis, an interactive dashboard was created using Power BI to visually present the insights.

This project demonstrates practical skills in data preprocessing, exploratory data analysis (EDA), feature engineering, and data visualization.

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**2. Dataset Information**

The dataset used for this project was obtained from Kaggle and contains detailed information about Netflix movies and TV shows.

The dataset includes the following attributes:

- Title of the content
- Type (Movie or TV Show)
- Genre / Listed In
- Country of origin
- Release Year
- Rating category
- Duration
- Date added to Netflix
- Description

This dataset provides valuable information to analyze the content trends of Netflix and understand how the platform's content library has evolved over time.

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**3. Tools and Technologies Used**

The following tools and technologies were used in this project:

- Python
- Pandas
- Jupyter Lab
- Power BI

- Data Visualization Techniques

Python and Pandas were used for data cleaning and transformation, while Power BI was used to create an interactive dashboard to visualize the results.

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**4. Data Cleaning and Preprocessing**

Before performing the analysis, several data cleaning and preprocessing steps were carried out to ensure the dataset was accurate and suitable for analysis.

The following operations were performed:

- Checked the dataset structure using Pandas functions such as info() and describe().
- Identified and handled missing values using isnull() and appropriate data treatment methods.
- Removed or handled duplicate records to maintain data integrity.
- Converted the date_added column to datetime format to enable time-based analysis.
- Standardized text-based columns to maintain consistency.
- Split multi-value columns such as genres and countries.
- Used the **explode() function** to separate multiple genres into individual rows for accurate genre analysis.
- Applied **groupby() operations** to aggregate data and perform statistical analysis across categories.
- Created new derived features using **feature engineering techniques**, such as extracting the year from the date column for trend analysis.

These preprocessing steps ensured that the dataset was structured and ready for further analysis and visualization.

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**5. Exploratory Data Analysis (EDA)**

After cleaning the dataset, exploratory data analysis was performed using Pandas to understand patterns and trends in the data.

Key analysis tasks included:

- Analyzing the distribution of **Movies vs TV Shows** on Netflix.
- Identifying the **most common genres** available on the platform.
- Analyzing **country-wise content distribution** to determine which countries contribute the most content.
- Examining the **average duration of movies and TV shows by genre**.
- Studying the **growth of Netflix content over the years**.

- Understanding the **rating categories and audience targeting**.

These analyses helped uncover patterns in Netflix's content library and provided valuable insights for visualization.

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**6. Data Visualization using Power BI**

After completing the analysis in Python, the processed dataset was imported into Power BI to build an interactive dashboard.

The dashboard includes multiple visualizations such as:

- Movies vs TV Shows distribution chart
- Genre distribution visualization
- Country-wise content analysis
- Average duration by genre
- Content growth trend over the years

The Power BI dashboard allows users to interact with the data through filters and slicers, making it easier to explore different aspects of Netflix content.

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**7. Key Insights**

Several meaningful insights were derived from the analysis:

- Movies represent a larger portion of Netflix content compared to TV Shows.
- Certain genres such as Drama and International Movies dominate the platform.
- The United States contributes the highest number of titles to Netflix.
- Netflix content production significantly increased after 2016, indicating rapid platform expansion.
- Genre-based analysis reveals variations in average movie durations.

These insights highlight the content trends and production patterns of Netflix over time.

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**8. Conclusion**

This project demonstrates how data analysis techniques can be applied to real-world datasets to extract meaningful insights. By using Python and Pandas for data cleaning and analysis, and Power BI for visualization, the project effectively transforms raw data into actionable insights.

The interactive dashboard provides a clear and structured overview of Netflix content trends, making it easier to understand the distribution of movies, genres, and global contributions.

This project also showcases essential data analyst skills including data preprocessing, exploratory data analysis, feature engineering, and dashboard creation.